

Rough Volatility: From Empirical Evidence to Pricing

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The Classic Heston Model

$$\frac{dS_t}{S_t} = \mu dt + \sqrt{V_t} dW_t^S$$
$$dV_t = \kappa(\theta - V_t)dt + \xi \sqrt{V_t} dW_t^V$$

with $d\langle W^S, W^V \rangle_t = \rho dt$.

Strengths :

- ▶ **Affine & Markovian:** Highly tractable mathematical properties.
- ▶ **Analytical pricing:** Semi-closed form solutions available via the Moment Generating Function.

Visualizing the Log Volatility of the S&P500 and the Log CIR Process

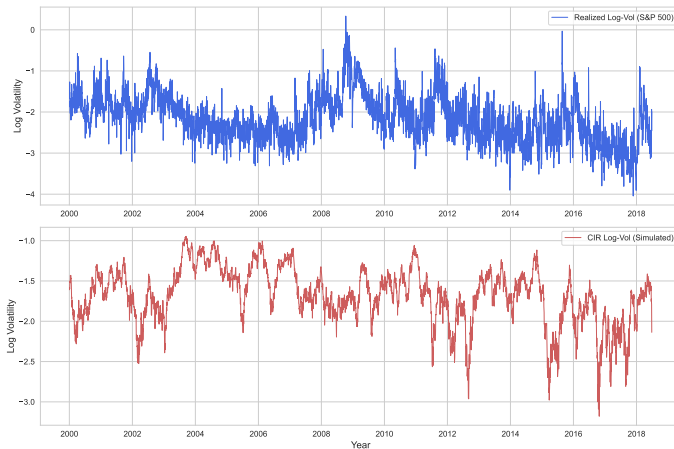


Figure: Daily S&P500 log-volatility vs Simulated Log CIR Process

The short-term skew anomaly

Let $k = \ln(K/S_0)$ be the log-moneyness.

Market observation

- ▶ Empirically, the short-term ATM skew explodes following a power law $T^{H-1/2}$:

$$\lim_{T \rightarrow 0} \left| \frac{\partial \sigma_{imp}}{\partial k} \right|_{k=0} = \infty$$

The Heston model failure

- ▶ In classical continuous stochastic volatility models, the short-term skew is structurally bounded.
- ▶ For the Heston model, asymptotic expansions show the skew converges to a **finite constant** defined by its parameters:

$$\lim_{T \rightarrow 0} \left| \frac{\partial \sigma_{imp}}{\partial k} \right|_{k=0}^{\text{Heston}} = \frac{\rho \xi}{4 V_0} < \infty$$

Volatility is Rough (Gatheral et al. 2018)

- 1 Volatility is Rough (Gatheral et al. 2018)
- 2 Pricing Under Rough Volatility (Bayer et al. 2016)
- 3 Turbocharging Monte Carlo pricing for the rough Bergomi model (McCrickerd et al. 2018)
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Fractional Brownian Motion: Core Properties

The fractional Brownian motion W_t^H is a centered, continuous Gaussian process governed by the Hurst exponent $H \in (0, 1)$.

Fundamental statistics:

- ▶ **Stationary increments:** $W_{t+\Delta}^H - W_t^H \stackrel{d}{=} W_\Delta^H$
- ▶ **Autocovariance:** $\mathbb{E}[W_t^H W_s^H] = \frac{1}{2}(t^{2H} + s^{2H} - |t - s|^{2H})$
- ▶ **Scaling:** $\mathbb{E}[|W_{t+\Delta}^H - W_t^H|^q] = K_q \Delta^{qH}$

Dependence on H :

- ▶ $H = 1/2$: Standard BM (independent increments).
- ▶ $H > 1/2$: Persistent paths.
- ▶ $H < 1/2$: Anti-persistent (rough) paths.

Visualizing fBm: The Diffusive Regime ($H = 0.5$)

Standard Brownian Motion

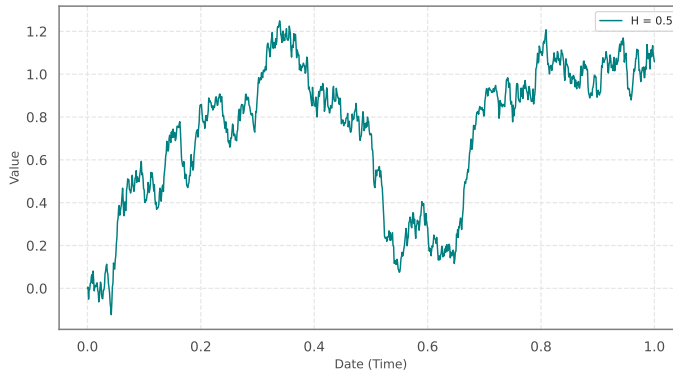


Figure: Simulated path of a standard Brownian motion ($H = 0.5$) with independent increments.

Visualizing fBm: The Persistent Regime ($H = 0.9$)

Trending / Long Memory Behavior

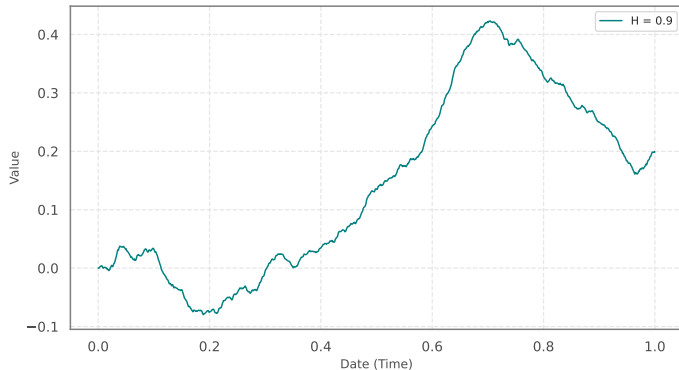


Figure: Simulated path of a fractional Brownian motion ($H = 0.9$), illustrating smooth trajectories.

Visualizing fBm: The Rough Regime ($H = 0.01$)

Anti-persistent / Mean-Reverting Behavior

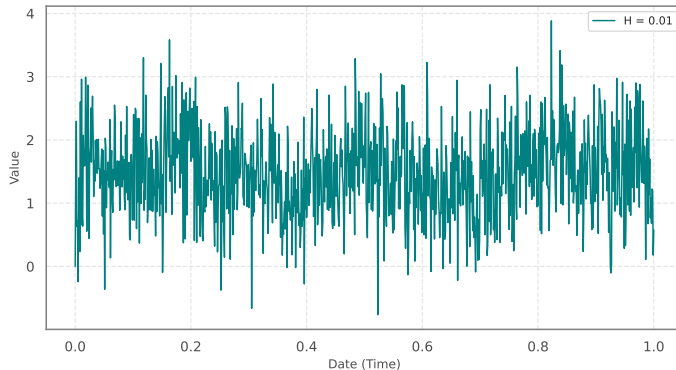


Figure: Simulated path of a fractional Brownian motion ($H = 0.01$), illustrating extreme roughness.

fBm Representations

Mandelbrot-Van Ness

$$W_t^H = \frac{1}{\Gamma(H + 1/2)} \left(\int_{-\infty}^t (t-s)^{H-1/2} dW_s - \int_{-\infty}^0 (-s)^{H-1/2} dW_s \right)$$

- ▶ Integrates over $(-\infty, t]$.
- ▶ Strictly guarantees stationary increments.

Volterra process (Riemann-Liouville)

$$\tilde{W}_t^H = \sqrt{2H} \int_0^t (t-s)^{H-1/2} dW_s$$

- ▶ History truncated at $t = 0$ (computationally tractable).
- ▶ Sacrifices strict stationarity but preserves exact **regularity**.

Extracting the Hurst exponent

Assuming log-volatility follows an fBm, we compute the empirical moments of its increments over a sample of N observations:

$$m(q, \Delta) := \frac{1}{N} \sum_{k=1}^N |\log(\sigma_{k\Delta}) - \log(\sigma_{(k-1)\Delta})|^q \approx \mathbb{E}[|\log(\sigma_\Delta) - \log(\sigma_0)|^q] = K_q \Delta^{qH}$$

Applying a logarithm yields a linear equation:

$$\log(m(q, \Delta)) \approx qH \log(\Delta) + \log(K_q)$$

By regressing $\log(m(q, \Delta))$ against $\log(\Delta)$, the slope directly reveals qH .

The Scaling Moments

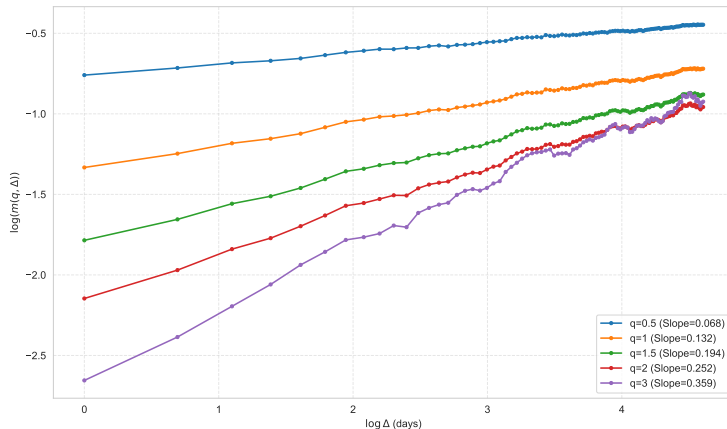


Figure: SPX empirical log-moments: strict linearity across q confirms monofractality with $H \approx 0.13$.

Pricing Under Rough Volatility (Bayer et al. 2016)

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The Classic Bergomi Model

Inspired by the HJM framework, Bergomi (2005) directly models the **forward variance curve** $\xi_t(u) = \mathbb{E}[\sigma_u^2 \mid \mathcal{F}_t]$, guaranteeing a perfect fit to the initial variance swap term structure.

General dynamics:

$$\begin{aligned}\frac{dS_t}{S_t} &= \sqrt{V_t} dZ_t \quad \text{with} \quad V_t = \xi_t(t) \\ \frac{d\xi_t(u)}{\xi_t(u)} &= \omega(t, u) dW_t\end{aligned}$$

The original specification:

- ▶ In the original paper, Bergomi uses an exponential kernel $\omega(t, u) = \nu e^{-\kappa(u-t)}$ to preserve a **Markovian structure** for tractable pricing.
- ▶ **Limitation:** This smooth, bounded kernel structurally prevents the model from capturing the explosive short-term skew.

The rough Bergomi (rBergomi) Model

$$\frac{dS_t}{S_t} = \sqrt{V_t} dZ_t \quad \text{with} \quad V_t = \xi_t(t)$$
$$\frac{d\xi_t(u)}{\xi_t(u)} = \eta \sqrt{2H} (u - t)^{H-1/2} dW_t$$

This fractional kernel makes the forward variance strictly **non-Markovian**. Integrating these dynamics yields the explicit process for the spot variance:

$$S_t = S_0 \mathcal{E} \left(\int_0^t \sqrt{V_u} dZ_u \right)$$
$$V_t = \xi_0(t) \mathcal{E} \left(\eta \sqrt{2H} \int_0^t (t - s)^{H-1/2} dW_s \right) = \xi_0(t) \mathcal{E} \left(\eta \tilde{W}_t^H \right)$$

where $\mathcal{E}(X)_t = \exp \left(X_t - \frac{1}{2} \langle X \rangle_t \right)$ denotes the stochastic exponential.

Exact simulation via Cholesky

The Gaussian foundation Because the Volterra process $\tilde{W}^H$ and the asset driver Z are **jointly Gaussian**, the system is entirely defined by its covariance structure:

$$\text{Autocovariance: } \mathbb{E}[\tilde{W}_t^H \tilde{W}_s^H] \quad \text{and} \quad \text{Cross-covariance: } \mathbb{E}[\tilde{W}_t^H Z_s]$$

Path generation strategy On a time grid of N steps, we compute the dense $2N \times 2N$ joint covariance matrix Σ .

Using its Cholesky decomposition $\Sigma = LL^\top$, we generate exact correlated paths directly from independent standard normals:

$$\begin{pmatrix} \tilde{W}^H \\ Z \end{pmatrix} = L \times \mathcal{N}(0, I_{2N})$$

Simulating the Dynamics

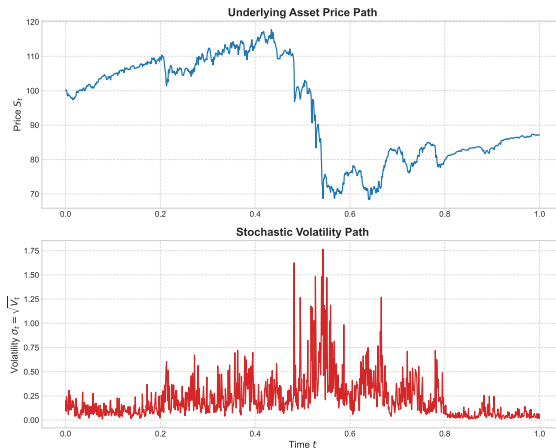
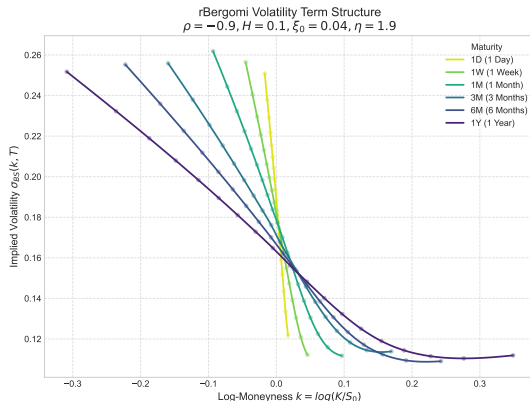


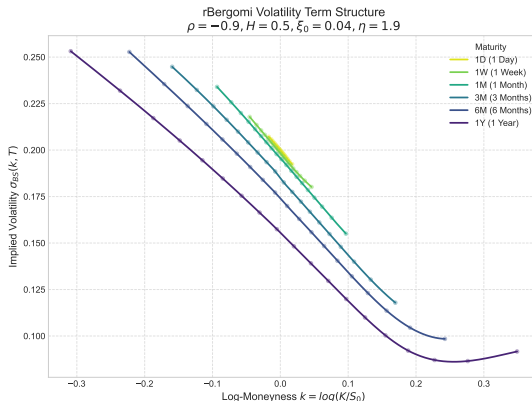
Figure: Simulation of the rBergomi asset price and stochastic volatility paths generated via Cholesky.

Term structure: The rough regime ($H = 0.1$)



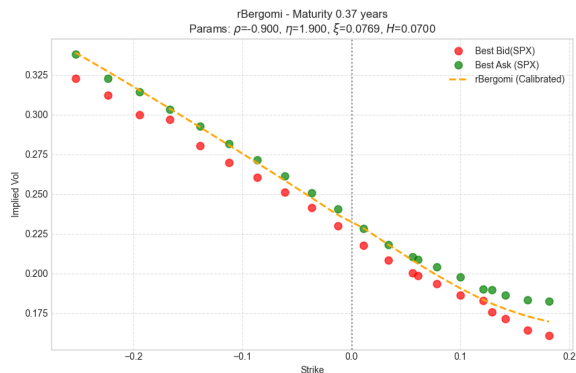
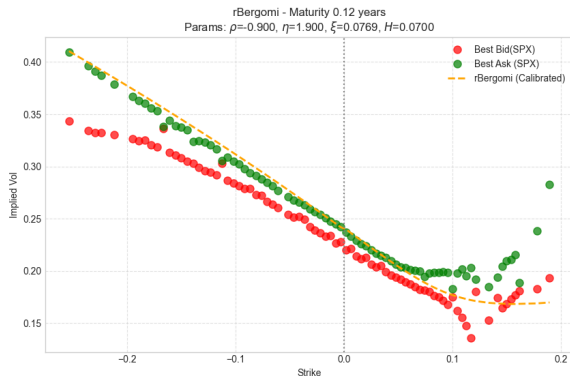
Simulated rBergomi volatility surface with $H = 0.1$. The non-Markovian fractional kernel naturally forces the ATM skew to explode as maturity $T \rightarrow 0$, matching empirical data.

Term structure: The classical regime ($H = 0.5$)



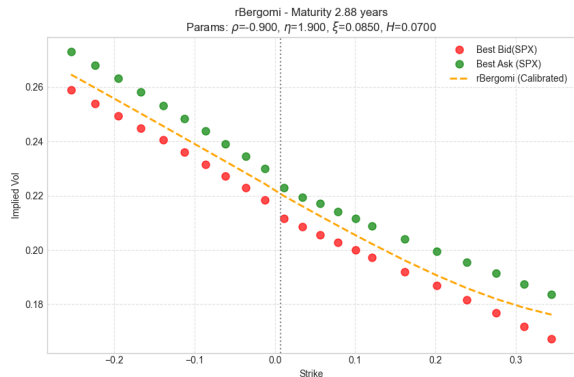
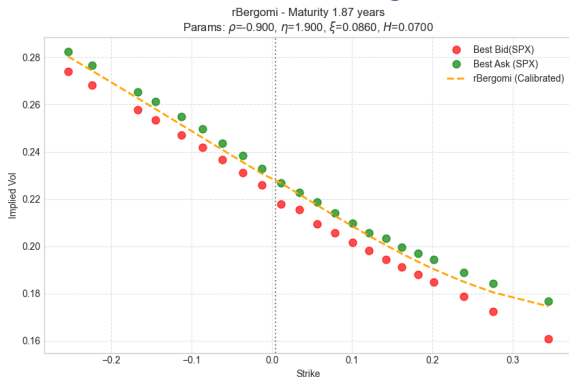
Simulated rBergomi volatility surface with $H = 0.5$. The model reduces to a standard diffusion where the short-term skew remains structurally bounded, failing to steepen.

Heuristic surface fit: Short-term maturities



Short-term options ($T < 6$ months). Using only three guessed parameters ($H = 0.07, \eta = 1.9, \rho = -0.9$), the rBergomi model perfectly captures the explosive ATM skew.

Heuristic surface fit: Long-term maturities



Long-term options ($T \geq 6$ months). The exact same heuristic parameters maintain a robust structural fit across the surface, effectively balancing curvature and variance levels.

The computational cost

Breaking the Markov property

- ▶ Because the forward variance depends on the entire history of the Brownian motion, we lose closed-form formulas and PDEs.
- ▶ Pricing relies exclusively on Monte Carlo simulation.

The $\mathcal{O}(N^3)$ bottleneck

- ▶ Exact simulation of the correlated Volterra process via Cholesky decomposition requires building and inverting a dense joint covariance matrix.
- ▶ This makes classical calibration far too slow for practical, daily trading purposes.

Turbocharging Monte Carlo pricing for the rough Bergomi model (McCrickerd et al. 2018)

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The Hybrid Scheme and FFT acceleration

To bypass Cholesky, we approximate the Volterra process using the Hybrid Scheme. The integral is split into a singular recent part and a historical Riemann sum:

$$\tilde{W}_{\frac{i}{n}}^{\alpha} = \sqrt{2H} \left(\int_{\frac{i-1}{n}}^{\frac{i}{n}} \left(\frac{i}{n} - u \right)^{H-1/2} dW_u^1 + \sum_{k=2}^i \left(\frac{b_k}{n} \right)^{H-1/2} \Delta W_{\frac{i-k}{n}}^1 \right)$$

- ▶ The summation term natively takes the form of a **discrete convolution**.
- ▶ By evaluating it with the Fast Fourier Transform (FFT), simulation complexity collapses from $\mathcal{O}(N^3)$ to $\mathcal{O}(N \log N)$.

Performance Hybrid Scheme vs Cholesky



Benchmark of execution times. The FFT-powered Hybrid Scheme solves the path-generation bottleneck inherent to rough volatility models.

Variance reduction: The Mixed Estimator

We combine Conditional Monte Carlo with a Control Variate to form the mixed estimator:

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n (X_i + \hat{\alpha}_n Y_i) - \hat{\alpha}_n \mathbb{E}[Y]$$

1. Conditional Monte Carlo (X)

$$X = BS \left((1 - \rho^2) \int_0^t V_u du; S_t^1, k \right)$$

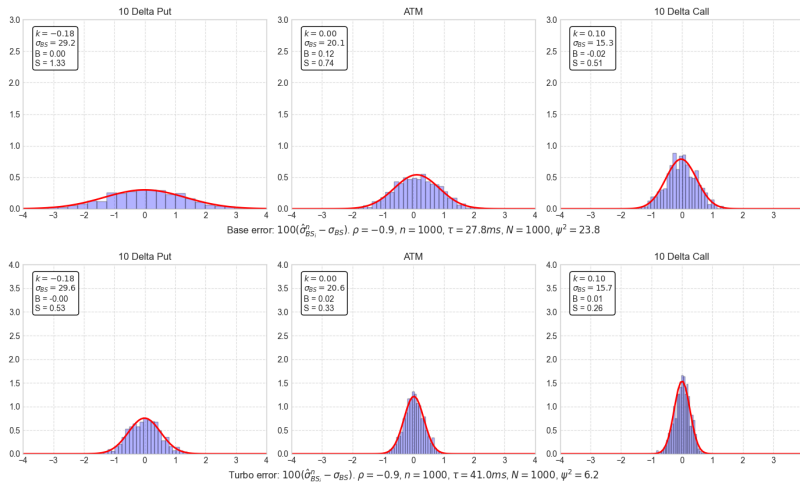
→ Integrates out the orthogonal price noise W^2 analytically via Black-Scholes.

2. Control Variate (Y)

$$Y = BS \left(\rho^2 \left(\hat{Q}_n - \int_0^t V_u du \right); S_t^1, k \right)$$

→ A synthetic Timer option with a known $\mathbb{E}[Y]$ that anchors paths and corrects sampling error.

Empirical variance reduction



Error distributions: Base estimator (top) vs. Mixed estimator (bottom).

S&P 500 Surface Calibration (Sept 16, 2022)

Optimal Parameters

$$H = 0.1464 \quad \eta = 1.7773 \quad \rho = -0.5797 \quad \xi_0 = 0.0688$$

$$\text{Calibration Error: RMSE} = 0.0126$$

Empirical validation

- ▶ The optimizer naturally converges to $H \approx 0.146$, perfectly aligning with our initial model-free measurements of market roughness!

Performance & Limits

- ▶ **Speed:** Optimization completes in ~ 5 minutes thanks to the mixed estimator.
- ▶ **Parsimony:** Highly accurate fit despite the restrictive assumption of a flat ξ_0 .
- ▶ *Note:* Bootstrapping a time-dependent curve $\xi_0(t)$ would further improve the fit for longer maturities.

Conclusion

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Thank you

Any questions?

References

- ▶ Gatheral, J., Jaisson, T., & Rosenbaum, M. (2018). *Volatility is rough*. Quantitative Finance, 18(6), 933-949.
- ▶ Bayer, C., Friz, P., & Gatheral, J. (2016). *Pricing under rough volatility*. Quantitative Finance, 16(6), 887-904.
- ▶ McCrickerd, R., & Pakkanen, M. S. (2018). *Turbocharging Monte Carlo pricing for the rough Bergomi model*. Quantitative Finance, 18(11), 1877-1886.